12. AI Tools for Engineers

Choosing, using, and verifying

Week 12 · CE/ARCE 2XX AI Literacy for Civil and Architectural Engineering



COLLEGE OF ENGINEERING
Civil & Architectural Engineering & Mechanics

Agenda

Today's plan

1. Quiz on last week (Week 11)
2. The AI tool landscape for engineers
3. Choosing a tool: fit, accuracy, privacy, cost, licensing
4. Trust, but verify: the 5-check workflow
5. AI for coding, and documenting AI use
6. Lab 12 preview, HW5, and project feedback

Last week in 60 seconds

Week 11 recap: AI across CE and ARCE

1 Every life-cycle stage

Plan, design, build, operate, maintain, renew — AI shows up in each one.

2 Every sub-discipline

From structural health monitoring to flood forecasting, site safety, and HVAC fault detection.

3 Our data is hard

Small data sets, rare failures, long lifespans, safety-critical decisions, codes and liability.

4 Lab 11 lesson

On imbalanced bridge data, accuracy misleads. Check recall for the class you care about.

Learning objectives

By the end of today, you can…

- ✓ Describe the main **categories** of AI tools and what each is good and bad at
- ✓ Choose a tool using **task fit, accuracy, privacy, cost, and licensing**
- ✓ Apply the **5-check verification workflow** to an AI answer and catch its errors
- ✓ Test AI-written code with known cases and `assert` before trusting it
- ✓ Write a clear **AI-use acknowledgment** statement

01

The landscape

Kinds of AI tools — categories, not brands

The landscape

Eight kinds of AI tools engineers meet (as of 2026)

Many products combine several of these — and features change fast

General chat assistants

Explain, draft, brainstorm, summarize, do quick math (verify it).

Coding assistants

Write, explain, and debug code inside an editor or notebook.

Research & search assistants

Search the web or papers and answer with citations — which you must open.

Document & spreadsheet AI

Summarize PDFs, draft reports, write formulas inside office software.

Image & visual tools

Generate concept images; read photos, sketches, and charts.

Domain tools

AI features inside BIM, CAD, structural, and energy software.

Local & open models

Run on your own computer: data stays local, but small models make more mistakes.

Agents

Tools that browse, run code, or edit files on their own (Week 10).

Match the task to the category — and what to check

Task	Category	What to check
Explain hydraulic radius before a quiz	Chat assistant	Compare with your textbook
Write Python to plot traffic counts	Coding assistant	Read it, run it, test with known values
Find studies on cool pavements in Phoenix	Research assistant	Open every citation; does it say that?
Summarize a 150-page geotechnical report	Document AI	Spot-check pages; is the report confidential?
Concept images for a studio pin-up	Image tool	A mood picture, not a design; usage rights
Auto-tag rooms in a BIM model	Domain tool	Vendor documentation; review a sample of tags

02

Choosing a tool

Five questions before you start

Choosing a tool

Five questions to ask before you use an AI tool

Task fit

Is this the right kind of tool? A spreadsheet or a formula may be better.

Accuracy

How good is it on YOUR kind of problem? Test it on cases you know.

Privacy

What data will it see, store, or train on? Is the tool approved for that data?

Cost

Subscription or per-use fees — plus your time to check the output.

Licensing & IP

Terms of use, who may use the output, open-model licenses, image rights.

Choosing a tool

Classify the data before you paste it

	Examples	Where it may go
Public	Published codes, textbook problems, course datasets, public reports	Most tools are fine — still verify
Internal	Your drafts, unpublished class or firm work, internal notes	University- or firm-approved tools only
Confidential (client)	Client drawings, reports under an NDA, bid prices	Only if the contract and policy allow; often: don't
Restricted	Personal data (student or worker records, health) and security-sensitive infrastructure details	Do not paste. Follow policy

Concept check: the NDA report

Your internship firm has a client's geotechnical report under an NDA. You want a quick summary. What should you do first?

A Paste it into a free chatbot — a summary is harmless

B Delete the client's name, then paste it into any tool

C Check the firm's AI policy and the NDA; use only an approved tool — or summarize it yourself

D Ask the chatbot to promise to keep the report confidential

Concept check: the NDA report — answer

Your internship firm has a client's geotechnical report under an NDA. You want a quick summary. What should you do first?

A Paste it into a free chatbot — a summary is harmless

B Delete the client's name, then paste it into any tool

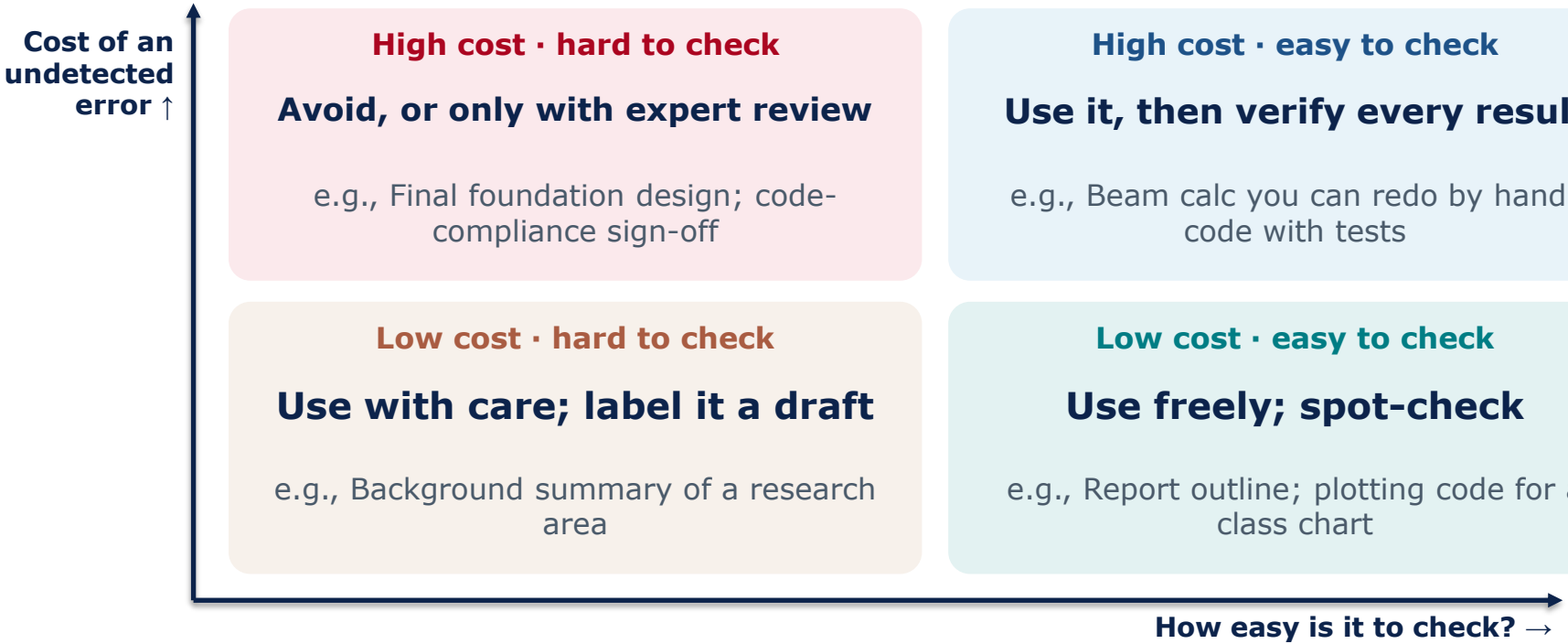
C Check the firm's AI policy and the NDA; use only an approved tool — or summarize it yourself

D Ask the chatbot to promise to keep the report confidential

Removing a name does not remove the obligation — location and project details still identify the client. The contract and the firm's policy decide.

Choosing a tool

Match how much you rely on AI to the risk



Pick the category, the data level, and the check

5 min

Think · Pair · Share

- **(a)** Summarize a city's published stormwater manual for a class project
- **(b)** Fix a bug in your Lab 11 notebook
- **(c)** Draft an email to a client about a delayed submittal, quoting the contract
- **(d)** Estimate HVAC loads for a real building your internship firm is designing

For each: which **tool category**? What **data level** (Public, Internal, Confidential, Restricted)? Which **check** matters most?

03

Trust, but verify

The 5-check verification workflow

Trust, but verify

AI makes a few kinds of errors again and again

Kind of error	Engineering example	Check that catches it
Wrong problem or assumption	Cantilever formula used for a simply supported beam	1 · Restate
Unit or decimal slip	Slope of 0.5 % entered as 0.5	2 · Units & magnitude
Arithmetic or formula slip	9.81 × 2 × 1000 reported as 1,962 Pa	3 · Independent check
Invented or misquoted source	“OSHA allows 1:1 slopes in Type C soil”	4 · Sources
Buggy code	Manning function uses R^(3/2) instead of R^(2/3)	5 · Test & document

Trust, but verify

The 5-check verification workflow

1	**Restate**	Restate the problem, inputs, and assumptions. Did the AI solve **your** problem?
2	**Units & magnitude**	Check units at every step, and whether the size of the answer is reasonable.
3	**Independent check**	Redo it by hand, in Python, or with a different method. Do the answers agree?
4	**Sources**	Confirm that cited codes and papers exist — and actually say that.
5	**Test & document**	Test generated code with known cases, and document your AI use.

Use these five names in Lab 12, HW5, and your project

Worked example 1

A confident answer with the wrong formula

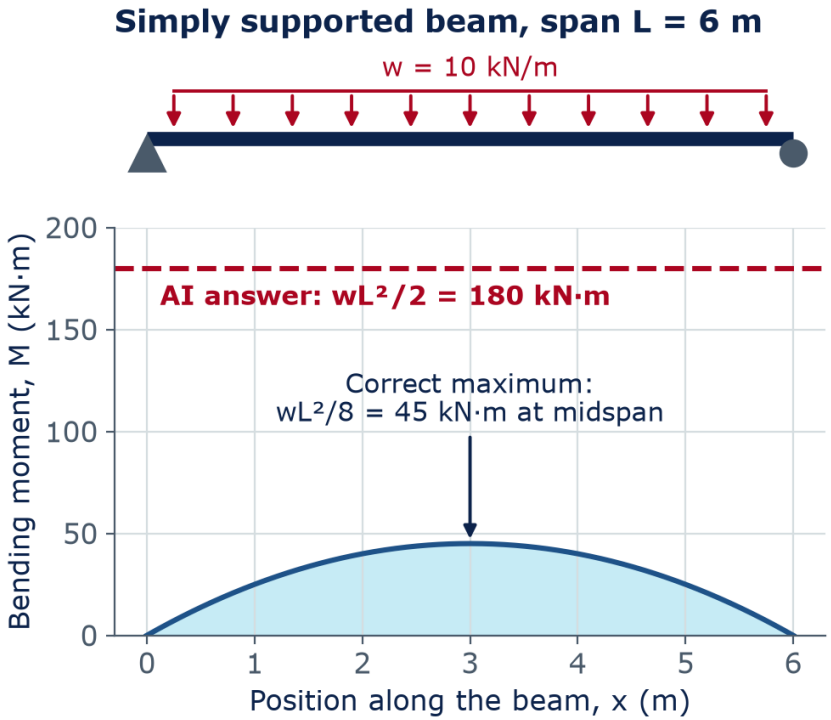
Problem: simply supported beam, span 6 m, uniform load w = 10 kN/m. Maximum moment?

AI answer: $M = wL^2/2 = 10 \times 6^2/2 =$ **180 kN·m**

Check 1 · Restate: "simply supported" — but $wL^2/2$ is the **cantilever** formula.

Check 3 · Independent check (statics):

$R = wL/2 = 30$ kN; $M(3\text{ m}) = 30\times3 - 10\times3\times1.5 =$ **45 kN·m** $= wL^2/8$



Worked example 2

Manning's equation gives open-channel flow

$$Q = \frac{1}{n} A R^{2/3} S^{1/2}$$

Q flow rate (m^3/s)

n Manning roughness — about 0.013 for finished concrete

A flow area (m^2)

R hydraulic radius = A / wetted perimeter P (m)

S channel slope (m/m) — a **decimal**, not a percent

Worked example 2

A percent slipped in as a decimal

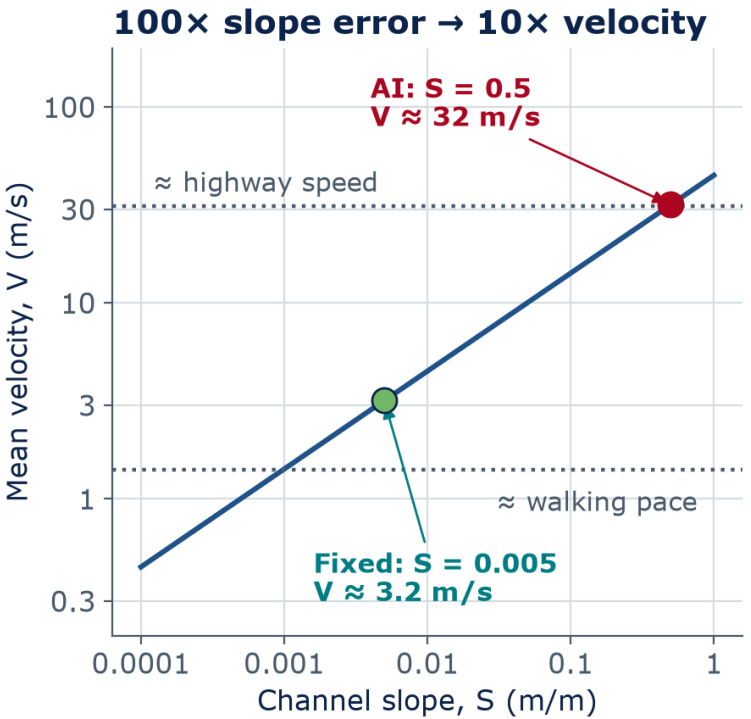
Problem: rectangular concrete channel, 2.0 m wide, flow depth 0.8 m, n = 0.013, slope 0.5 %. Find Q.

AI answer: A = 1.6 m², R = 0.444 m, S = **0.5** → Q ≈ 51 m³/s

Check 2 · Units & magnitude:

V = Q/A = 51 / 1.6 ≈ **32 m/s** (≈ 114 km/h) — highway speed in a small channel? No.

Fix: 0.5 % = 0.005 → Q ≈ **5.1 m³/s**, V ≈ 3.2 m/s



Trust, but verify

Concept check: which check catches it?

An AI says: “For Type C soil in a 12 ft deep excavation, OSHA allows a maximum slope of 1:1 (45°).” Which check catches this most directly?

- **A** 1 · Restate
- **B** 2 · Units & magnitude
- **C** 4 · Sources
- **D** 5 · Test & document

Concept check: which check catches it? — answer

An AI says: “For Type C soil in a 12 ft deep excavation, OSHA allows a maximum slope of 1:1 (45°).” Which check catches this most directly?

- **A** 1 · Restate
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Table B-1 (29 CFR 1926 Subpart P, App. B) gives 1½H:1V (34°) for Type C. 1:1 is the Type B value — a real number attached to the wrong soil.

Trust, but verify

Check 4, Sources: four questions for every citation

1 Does it exist?

Find it yourself: library, publisher, eCFR. Invented citations look real.

2 Right version?

Codes change by edition (ASCE 7-16 vs 7-22), and adoption varies by city.

3 Does it say that?

Read the section. AI often attaches a real source to a wrong claim.

4 Primary source?

Prefer the code, standard, or paper over a blog or summary.

Trust, but verify

Quick check: catch the slips

3 min

Quick calculation

- **AI answer 1:** a slab 10 m × 8 m × 150 mm thick needs **1,200 m³** of concrete
- **AI answer 2:** water pressure at the bottom of a 2 m deep tank is **2 kPa**
- Which check flags each one? What is the right answer?

Hint: about how many truckloads is 1,200 m³? How many kPa per meter of water depth?

04

AI for coding, and documenting AI use

Read before you run; test before you trust

AI for coding

Read before you run

1 Read every line

Can you explain what each line does? If not, ask — then check the explanation.

2 Check units and inputs

Names, units, and assumptions should be stated in comments and docstrings.

3 Never paste secrets

No passwords, API keys, or client data in prompts or shared code.

4 Check the imports

AI sometimes names packages that do not exist; attackers have registered such names.

AI for coding (lecture-only example)

Test generated code with known cases

```
# AI-written: peak runoff, rational method (SI)
def peak_runoff(C, i_mm_per_h, area_ha):
    """Peak flow Q in m^3/s."""
    return C * i_mm_per_h * area_ha / 3600

# Known case by hand: C=0.9, i=50 mm/h, A=2 ha
# gives Q = 0.25 m^3/s
assert abs(peak_runoff(0.9, 50, 2) - 0.25) < 1e-6
# -> AssertionError: it returns 0.025

# Edge case: no area, no runoff
assert peak_runoff(0.9, 50, 0) == 0
```

- `assert` stops the program if the condition is false
- Units: 1 mm/h on 1 ha = 10 m^3/h = 1/360 m^3/s
- 3600 looks right (seconds per hour) but skips mm → m and ha → m^2
- Fix: divide by **360** — and keep the test

Lecture-only example. Lab 12, Part 4 tests a different AI-written function (Manning's equation).

Live demo

Same tool, same question — two workflows

Copy-paste workflow

- Paste the question with no context
- Copy the first answer
- Trust the confident tone
- Submit — no record of AI use

vs

Verify workflow

- Restate the problem with inputs, units, assumptions
- Ask for steps, units, and stated assumptions
- Run the 5 checks; fix and re-ask
- Record tool, prompts, and what you checked

The tool is the same. The workflow decides whether the answer is usable.

An AI-use acknowledgment says what you used and how you checked it

Tool and version	Specific use	Prompts	How you checked
Which tool, which version	What you used it for	Paste or summarize them	What you verified, and how

Example (Lab 12, Part 4)

"I acknowledge the use of **[tool and version]** to draft the `manning_velocity` function in Part 4. Prompts I used included "Write a Python function for Manning's equation in SI units." I checked the output by writing two `assert` tests against hand calculations; they caught an exponent error (R^(3/2) instead of R^(2/3)) that a test with R = 1 m had missed, and I fixed it."

Lab 12

Lab day: Trust but Verify

LAB

12

Bring a charged laptop

Open the notebook from the course GitHub page

- Ask a small LLM eight CE/ARCE questions: beam moment, unit conversions, Manning, wall heat loss `Q = U·A·ΔT`, w/c ratio, OSHA slope, trip generation, rational method
- Check every answer in Python with the 5 checks; tabulate correct vs incorrect by problem type
- Part 4: catch the bug in an AI-written Manning function with `assert` tests
- Exit ticket: 3 short questions you answer alone, without AI, in the last 10 minutes
- **Project check-in, wave 1:** half the teams, last 25 minutes

This week

HW5 is posted; project feedback is back

DUE END OF WEEK 14

HW5 — Responsible AI in Practice

- **A (AI-Free):** find the errors in an AI-generated engineering calculation
- **B (AI-Assisted):** audit a model for data shift and subgroup error
- **C (AI-Assisted + acknowledgment):** memo — should a firm adopt AI tool X for task Y?

PROPOSAL FEEDBACK RETURNED

Final project

- Read the feedback with your team; revise your canvas
- Check-in (Lab 12 or Lab 13 — see your wave): data loaded and a baseline running
- Use the 5 checks on your own results — verification is in the rubric

Wrap-up

Key takeaways

1. Think in tool **categories**, not brands — products change fast; the questions don't
2. Classify data before you paste; Confidential and Restricted data need approved tools or none
3. **Restate → Units & magnitude → Independent check → Sources → Test & document**
4. AI-written code is a draft: read it, test it with known cases, then use it
5. Document AI use — tool and version, use, prompts, how you checked. **You remain responsible.**

Learn more

Readings and resources

- **OSHA excavation rules, 29 CFR 1926 Subpart P (eCFR)** — the primary source behind Table B-1
- **Python documentation: the assert statement** — 2-minute read
- **NIST AI 600-1: Generative AI Profile (2024)** — skim Section 2, its list of 12 generative-AI risks (preview of next week)
- **U.S. Copyright Office: Copyright and Artificial Intelligence** — current U.S. guidance on who can own AI-assisted work (it changes — check the date)
- **AI-use guide (course resources folder)** — the acknowledgment template and the 5 checks
- **Lab 12 notebook (course labs folder)** — open it in Colab on lab day